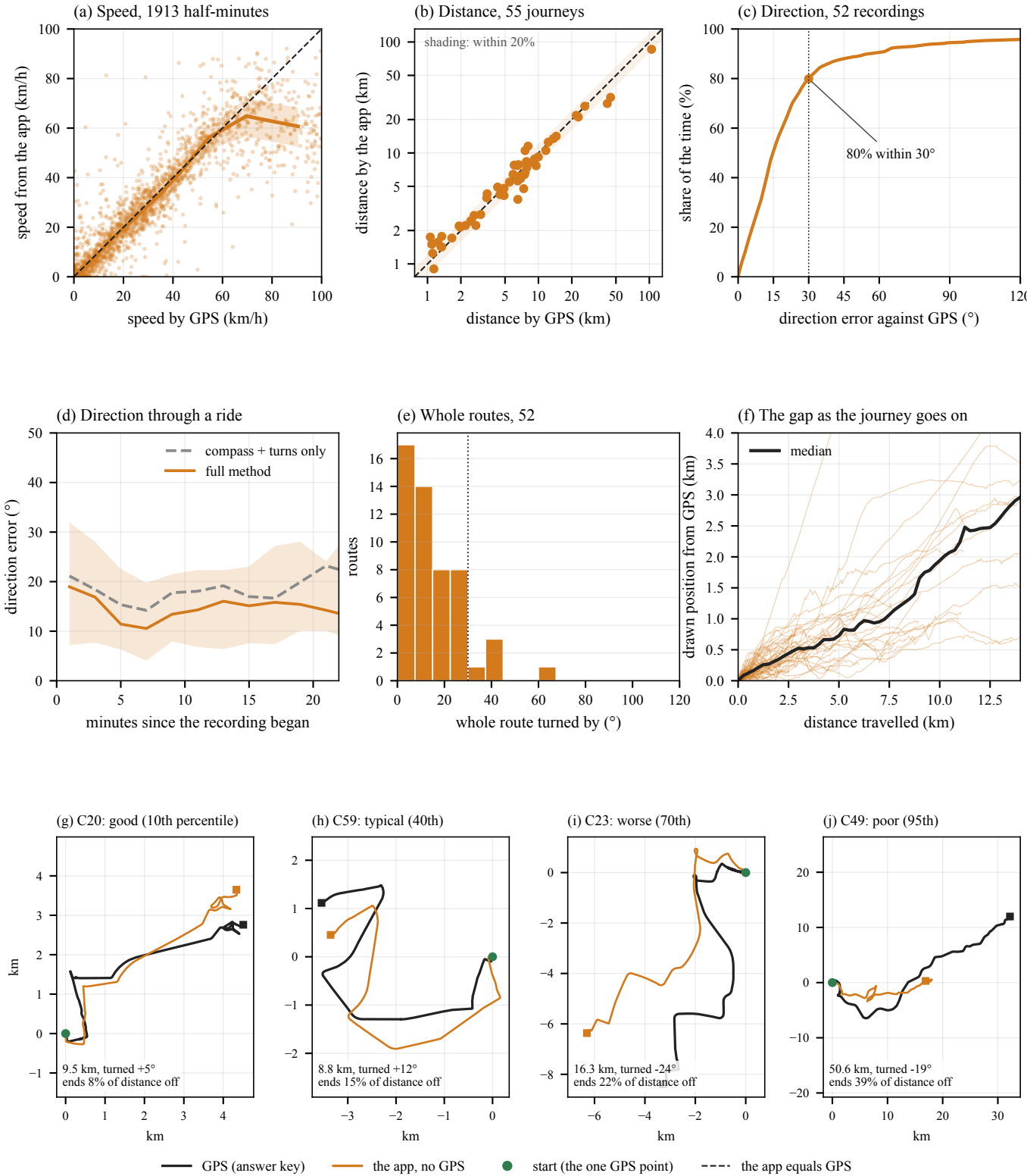

(a) Speed, 1913 half-minutes
speed from the app (km/h)
speed by GPS (km/h)
(b) Distance, 55 journeys
shading: within 20%
distance by the app (km)
distance by GPS (km)
(c) Direction, 52 recordings
share of the time (%)
80% within 30°
direction error against GPS (°)
(d) Direction through a ride
compass + turns only
full method
direction error (°)
minutes since the recording began
(e) Whole routes, 52
routes
whole route turned by (°)
(f) The gap as the journey goes on
median
drawn position from GPS (km)
distance travelled (km)
(g) C20: good (10th percentile)
9.5 km, turned +5°
ends 8% of distance off
(h) C59: typical (40th)
8.8 km, turned +12°
ends 15% of distance off
(i) C23: worse (70th)
16.3 km, turned -24°
ends 22% of distance off
(j) C49: poor (95th)
50.6 km, turned -19°
ends 39% of distance off
km
GPS (answer key)
the app, no GPS
start (the one GPS point)
the app equals GPS